

#Dataset

```
import pandas as pd

data = {
    'Customer_ID':[1,2,3,4,5,6,7,8,9,10],
    'Age':[22,35,45,28,32,50,41,29,38,26],

    'Gender':['Male','Female','Male','Female','Male','Female','Male','Female','Male','Female'],
    'Income':[25000,45000,60000,30000,50000,70000,65000,35000,55000,28000],

    'City':['Hyderabad','Mumbai','Delhi','Chennai','Bangalore','Delhi','Mumbai','Hyderabad','Bangalore','Chennai']
}

df = pd.DataFrame(data)

df.head()
```

#Data Cleaning

```
print(df.isnull().sum())

df.drop_duplicates(inplace=True)

print("Data Cleaned Successfully")
```

#Age Distribution Graph

```
import matplotlib.pyplot as plt

plt.figure(figsize=(8,5))
plt.hist(df['Age'], bins=5)

plt.title("Customer Age Distribution")
plt.xlabel("Age")
plt.ylabel("Number of Customers")

plt.show()
```

#Gender Distribution Graph

```
gender_count = df['Gender'].value_counts()

plt.figure(figsize=(6,6))
plt.pie(gender_count,
        labels=gender_count.index,
        autopct='%1.1f%%')

plt.title("Gender Distribution")
plt.show()
```

#Income Analysis

```
plt.figure(figsize=(8,5))
plt.bar(df['Customer_ID'], df['Income'])
```

```
plt.title("Customer Income Analysis")
plt.xlabel("Customer ID")
plt.ylabel("Income")
```

```
plt.show()
```

#City-wise Customer Count

```
city_count = df['City'].value_counts()
```

```
plt.figure(figsize=(8,5))
plt.bar(city_count.index, city_count.values)
```

```
plt.title("Customers by City")
plt.xlabel("City")
plt.ylabel("Number of Customers")
plt.show()
```

#Prediction Model (Income Prediction Based on Age)

```
from sklearn.linear_model import LinearRegression
import numpy as np
```

```
X = df[['Age']]
y = df['Income']
```

```
model = LinearRegression()
model.fit(X,y)
```

```
predicted_income = model.predict([[30]])
```

```
print("Predicted Income for Age 30:", predicted_income[0])
```